Diss. ETH No. _____

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(Dr. sc. ETH Zurich)

presented by

Firstname Lastname
born on dd.mm.yyyy

accepted on the recommendation of

Prof. Dr. Supervisor Name, supervisor
Prof. Dr. Co-Examiner Name, co-examiner

2026

Acknowledgements

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Zurich, March 2026

Firstname Lastname

Abstract

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Zusammenfassung

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1 Introduction

The exponential growth in data-intensive applications — from machine learning inference to large-scale scientific computing — places unprecedented demands on the hardware architectures that support them. At the same time, the end of classical Dennard scaling has made energy efficiency a first-class design constraint alongside raw performance [1]. The resulting tension between performance, energy, and area (Power, Performance, and Area (PPA)) motivates the search for fundamentally new approaches to hardware design.

This thesis investigates the design and implementation of scalable interconnect architectures for Systems-on-Chip (SoCs). Interconnects — the networks that carry data between processing elements, memories, and peripherals — are increasingly the primary bottleneck in modern systems: they account for a substantial fraction of the total chip area and power budget, yet their design has received comparatively little attention relative to compute elements [2].

1.1 Motivation

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The Integrated Systems Laboratory (Integrated Systems Laboratory (IIS)) at ETH Zurich has a long history of producing open-source hardware that bridges the gap between academic research and industrial

practice [3]. This thesis continues that tradition by contributing a fully open-source, silicon-validated interconnect architecture.

1.2 Problem Statement

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Formally, we seek an interconnect architecture $\mathcal{N}$ that satisfies:

$$\text{throughput}(\mathcal{N}) \geq T_{\text{min}}, \quad \text{power}(\mathcal{N}) \leq P_{\text{max}}$$

subject to an area budget A_{max} and full protocol compliance with the AXI4 standard.

1.3 Contributions

The main contributions of this dissertation are:

1. **NoC micro-architecture:** A novel decoupled control/data path design that achieves near-ideal pipeline utilization with minimal area overhead.
2. **Formal verification methodology:** A composable correctness argument that covers all protocol states without exhaustive simulation.
3. **Silicon validation:** A 22 nm Application-Specific Integrated Circuit (ASIC) implementation demonstrating 2.1× throughput improvement and 35% power reduction over prior work.
4. **Open-source release:** All Register-Transfer Level (RTL), scripts, and testbenches are released under a permissive open-source license.

1.4 Thesis Outline

The remainder of this thesis is organized as follows. Chapter 2 introduces the necessary background on Network-on-Chips (NoCs) and SoCs. Chapter 3 summarizes the contributions and outlines directions for future work.

Table 1.1: Overview of the thesis structure.

Chapter	**Content**
Chapter 2	Background on interconnects and related work
Chapter 3	Conclusions and future work

2 Background

This chapter reviews the background knowledge required to understand the contributions of this thesis. We cover the fundamentals of SoCs interconnect design (Section 2.1), standard communication protocols (Section 2.2), and prior work that motivates our approach (Section 2.3).

2.1 Network-on-Chip Fundamentals

A NoC is a packet-switched communication fabric that connects multiple intellectual property (IP) blocks on a single die [2]. Unlike traditional bus-based interconnects, NoCs scale to large numbers of endpoints while providing predictable worst-case latency and high aggregate bandwidth.

2.1.1 Topology

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The choice of topology determines the fundamental trade-offs between bisection bandwidth, hop count, and router area. Common topologies include:

- **Mesh:** Regular 2D grid; simple to implement and route; used in most commercial many-core processors.
- **Torus:** Mesh with wrap-around links; reduces diameter at the cost of longer wires.
- **Butterfly / fat-tree:** Hierarchical structures that minimize hop count at the cost of non-uniform link widths.

2.1.2 Routing

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The routing algorithm determines which path a packet follows from source to destination. Dimension-ordered routing (XY routing) is the de facto standard for mesh topologies due to its deadlock-free, deterministic nature.

2.1.3 Flow Control

Flow control governs how buffer space is allocated along a packet's path. Credit-based flow control — where a receiver grants credits to its upstream neighbor — is widely used because it provides tight, cycle-accurate back-pressure with low overhead.

2.2 Communication Protocols

Modern SoCs rely on standardized bus protocols to ensure interoperability between IP blocks from different vendors. The AXI4 protocol from the ARM AMBA specification is the dominant standard in the industry today.

2.2.1 AXI4 Overview

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AXI4 separates each transaction into five independent channels: address write (AW), data write (W), write response (B), address read (AR), and data read (R). This separation allows requests and responses to be pipelined independently, enabling high-throughput operation even under heavy load.

2.2.2 AXI4 Stream

For bulk data transfers, AXI4-Stream provides a simplified unidirectional interface without address channels. It is widely used for DMA engines and streaming accelerators.

2.3 Related Work

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[3] presents the FlooNoC interconnect, which achieves 645 Gb/s/link with 0.15 pJ/B/hop in a 22 nm technology. This work builds directly on FlooNoC and extends its micro-architecture to support multi-stream parallel transfers.

Table 2.1: Comparison of on-chip interconnect designs.

Work	BW (Gb/s/link)	Energy (pJ/B)	Technology
Prior A	128	0.48	28 nm
Prior B	256	0.31	22 nm
FlooNoC [3]	645	0.15	22 nm
This work	**1280**	**0.09**	**22 nm**

As shown in Table 2.1, our design achieves the best energy efficiency while doubling the peak bandwidth of FlooNoC.

3 Conclusions

This dissertation presented a novel approach to on-chip interconnect design that addresses the growing performance and energy demands of modern SoCs. We demonstrated that by carefully co-designing the micro-architecture, the verification methodology, and the physical implementation, it is possible to simultaneously improve throughput, reduce energy consumption, and simplify correctness proofs.

3.1 Summary of Contributions

The key contributions of this work are as follows.

Chapter 2 — Background: We reviewed the state of the art in NoC design and identified the key limitations of existing approaches that motivate our research.

Interconnect micro-architecture: We proposed a decoupled control/data path architecture that enables near-ideal pipeline utilization. The design was validated through extensive simulation and subsequently taped out in a 22 nm process, confirming simulation predictions with less than 3% deviation.

Formal verification methodology: We developed a composable correctness argument for AXI4-compliant interconnects that covers all reachable protocol states. Compared to conventional simulation, our approach reduced the verification effort by 60% while achieving higher coverage.

Silicon results: The fabricated chip achieves 1280 Gb/s/link at 0.09 pJ/B/hop — a 2× throughput improvement and 40% energy reduction over the best prior work at the same technology node. The Instructions Per Cycle (IPC) of the attached processor cores improved by 18% on average across our benchmark suite.

3.2 Future Work

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Several promising directions remain for future investigation:

- **Multi-die integration:** Extending the architecture to support chiplet-based systems via die-to-die protocols such as UCIe.
- **Adaptive routing:** Incorporating load-adaptive routing to handle traffic hotspots that arise in irregular workloads.
- **Automated topology generation:** Developing tools that automatically select the optimal topology and link widths for a given application's communication graph.
- **Formal scalability:** Extending the composable verification framework to cover heterogeneous topologies and mixed-protocol endpoints.

We believe that the foundations laid in this dissertation — particularly the open-source RTL and the formal verification methodology — will serve as a productive basis for these future directions.

A AXI4 Protocol Reference

A.1 Channel Signals

Table 1.1 lists the signals on the AXI4 write-address channel. The remaining channels follow an analogous structure.

Table 1.1: AXI4 write-address channel (AW) signals.

Signal	Width	Description
`AWID`	varies	Write address ID
`AWADDR`	32/64 b	Write address
`AWLEN`	8 b	Burst length (number of transfers minus one)
`AWSIZE`	3 b	Transfer size (bytes per beat, log2)
`AWBURST`	2 b	Burst type: FIXED, INCR, WRAP
`AWVALID`	1 b	Handshake: master asserts when address is valid
`AWREADY`	1 b	Handshake: slave asserts when it can accept the address

A.2 Handshake Protocol

Data is transferred on a channel whenever both `VALID` and `READY` are asserted in the same clock cycle. The master must not wait for `READY` before asserting `VALID`, and the slave must not wait for `VALID` before asserting `READY`. This rule prevents deadlock in systems where `READY` depends combinationally on `VALID`.

B | Chip Gallery

This appendix lists all fabricated chips the author has contributed to in the form of technical work or supervision.

B.1 Chip A

Chip photo

Application	High-performance NoC / Research Project
Technology	TSMC Xnm / Custom Package
Dimensions	$X.X$ mm $\times$ $Y.Y$ mm
Gate count	Z MGE
Voltage	V V
Clock	F GHz

Designers
Firstname Lastname, Collaborator One, Collaborator Two, Supervisor Name

Replace this paragraph with a description of the chip: its purpose, architecture highlights, key collaborators or funding sources, and any measurement results or expected outcomes.

B.2 Chip B

Chip photo

Application	IoT Processor / Research Project
Technology	GF Xnm / QFN Package
Dimensions	$X.X$ mm $\times$ $Y.Y$ mm
Gate count	Z MGE
Voltage	V V – W V
Clock	F MHz

Designers
Collaborator One, **Firstname Lastname**, Collaborator Two, Supervisor Name

Replace this paragraph with a description of the chip.

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[1] J. L. Hennessy and D. A. Patterson, "A New Golden Age for Computer Architecture," *Communications of the ACM*, vol. 62, no. 2, pp. 48–60, 2019.

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List of Acronyms

ALU	Arithmetic Logic Unit
ASIC	Application-Specific Integrated Circuit
DUT	Design Under Test
EDA	Electronic Design Automation
FPGA	Field-Programmable Gate Array
HDL	Hardware Description Language
IC	Integrated Circuit
IIS	Integrated Systems Laboratory
IPC	Instructions Per Cycle
ISA	Instruction Set Architecture
LLC	Last-Level Cache
NoC	Network-on-Chip
PPA	Power, Performance, and Area
RTL	Register-Transfer Level
SoC	System-on-Chip
UVM	Universal Verification Methodology

Curriculum Vitae

Firstname Lastname was born on dd Month YYYY in Hometown, Country. He/She received the B.Sc. and M.Sc. degrees in [Field] from [University] in YYYY and YYYY, respectively. Subsequently, he/she joined the [Laboratory] at [University] under the supervision of Prof. Dr. [Supervisor] to pursue a doctoral degree. His/Her research interests include [topic A] and [topic B].

Photo

Education

20XX – 20XX	**Ph.D. in [Field]**, [University]. Dissertation: *"Title of This Dissertation"*. Supervisor: Prof. Dr. Supervisor Name.
20XX – 20XX	**M.Sc. in [Field]**, [University].
20XX – 20XX	**B.Sc. in [Field]**, [University].

Publications

Journal Articles

[1] F. Lastname, A. Coauthor, and P. Professor, "Title of Journal Paper," *IEEE Trans. VLSI Systems*, vol. XX, no. X, pp. 1–14, 20XX.

[2] F. Lastname and A. Coauthor, "Title of Second Journal Paper," *IEEE Design & Test*, vol. XX, no. X, pp. 1–10, 20XX.

Conference Papers

[1] F. Lastname and A. Coauthor, "Title of Conference Paper," in *Proc. Design Automation Conference (DAC)*, 20XX.

[2] F. Lastname and A. Coauthor, "Title of Second Conference Paper," in *Proc. DATE*, 20XX.

Chip Tape-outs

Name	**Technology**	**Year**	**Role**
Chip A	TSMC Xnm	20XX	Lead designer
Chip B	GF Xnm	20XX	Contributor

See Appendix B for details on each chip.

Teaching

20XX – 20XX	Head Teaching Assistant, *Course Name*, [University].
20XX	Teaching Assistant, *Course Name*, [University].

Awards & Honors

- Best Paper Award, Conference Name, 20XX.
- Excellence Scholarship, [University], 20XX.